

▶ INTERACTIVE VIDEO WALKTHROUGH

Full System Feature Showcase & Code Review

Watch the comprehensive demonstration covering all four applications, live interactions, authentication flows, database states, and architectural highlights without running local servers.

✓ 4 Full-Stack Demos ✓ Real DB Queries & Seeding ✓ Auth & Role Workflows

URL: <https://youtu.be/GS0Nik1QADQ?si=qM3gDQe1QtST0J2q>

Watch Video Demo ↗

1. Environment Prerequisites & Compatibility Matrix

Runtime & Package Manager Standard		Database & Cloud Services Atlas / Local	
Node.js	v20.x or v22.x LTS (Recommended: 20.18+)	MongoDB Atlas	Cluster v6.0+ (Free M0 tier or higher)
Package Manager	npm 10.x+ or pnpm 9.x+	Local MongoDB	mongodb://127.0.0.1:27017/<dbname>
Operating System	Ubuntu / Debian / macOS / Windows (WSL2 recommended)	Network Access	Whitelist IP 0.0.0.0/0 in Atlas Network Access

2. Universal Architecture & Security Invariants

Zero-Mock Data & Production Integrity Every view, calculation, chart, and status stepper communicates with real MongoDB database collections and server endpoints. No fabricated or mocked client states.	Argon2id + Dual-Cookie JWT Security Salted hashes via Argon2id (Memory: 64MB, Iterations: 3, Parallelism: 4), short-lived access tokens (15m), and rotating refresh tokens (7d) with replay detection.
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3. Universal 3-Step Project Execution

All four applications in this monorepo are completely decoupled with independent package configurations and models. Running any project follows this unified sequence:

```
# Step 1: Navigate into the chosen project directory
cd <project_name>      # expense_tracker, job_portal_application, etc.

# Step 2: Install dependencies (if not already installed)
npm install

# Step 3: Run the development server
npm run dev            # Launches at http://localhost:3000
```

4. Detailed Application Setup Guides (Projects 1 & 2)

1. Expense Tracker & Wealth Intelligence (`expense_tracker/`)

Default Port: 3000

Personal finance intelligence platform with integer minor-unit currency arithmetic (zero floating-point flaws), sub-10ms MongoDB aggregation pipelines, dynamic category budgets, and dual-cookie JWT session rotation.

```
cd expense_tracker
npm install
cp .env.example .env.local      # Contains pre-configured MongoDB & JWT secrets
npm run dev                     # Launches at http://localhost:3000
```

DEMO ACCOUNT	EMAIL	PASSWORD	PRELOADED CONTEXT
Interactive Demo	demo@auspify.com	DemoUser1234!	100+ historical transactions, active budget targets, financial metrics

2. Job Portal Application (`job_portal_application/`)

Default Port: 3000 / 3001

Paperfolio Neo-Brutalist ATS & talent marketplace featuring candidate profile snapshot preservation, compound unique application indices, employer candidate funnel, and administrator audit logging.

```
cd job_portal_application
npm install
cp .env.example .env.local
npm run db:index      # Sync compound unique database indexes
npm run db:seed       # Seed categories, demo jobs, companies, and accounts
npm run dev           # Launches at http://localhost:3000
```

ROLE	EMAIL	PASSWORD	PRELOADED CONTEXT
Job Seeker	seeker@demo.local	DemoPassword123!	Complete profile, submitted applications, saved jobs
Employer	employer@demo.local	DemoPassword123!	Verified company ("Vortex Systems"), 3 active listings, applicant funnel
Administrator	admin@demo.local	DemoPassword123!	Platform moderation, audit logging, user suspension safeguards

Pre-configured Databases: Pre-configured `.env.local` files connected to live MongoDB Atlas clusters are already included in each project directory. You can test and inspect the applications immediately without provisioning new database clusters.

5. Detailed Application Setup Guides (Projects 3 & 4)

3. Enterprise Learning Management System (`learning_management_system/`)

Default Port: 3000 / 3002

E-learning platform featuring server-authoritative timed quizzes, markdown/video curriculum player, instructor grading queues, and cryptographically verifiable digital certificates (`EDU-XXXXXXX`).

```
cd learning_management_system
npm install
cp .env.example .env.local
npm run db:seed           # Seed courses, interactive curriculum, and role accounts
npm run dev               # Launches at http://localhost:3000
```

ROLE	EMAIL	PASSWORD	PRELOADED CONTEXT
Student	student@lms.local	StudentDemo123!	Active enrollments, quiz evaluations, certificates
Instructor	instructor@lms.local	InstructorDemo123!	Curriculum studio, assignment authoring, grading queue
Administrator	admin@lms.local	SuperSecretAdminPassword123!	Course publication moderation, audit trail, user role control

4. Student Management System (`student_management_system/`)

Default Port: 3000 / 3003

Educational institution administrative application with student directory PII masking, homeroom cohort capacity limits, typed deletion safeguards, and formula-injection safe CSV exports.

```
cd student_management_system
npm install
cp .env.example .env.local
npm run db:index          # Synchronize database indexes
npm run db:seed           # Seed cohorts, students, and staff accounts
npm run dev               # Launches at http://localhost:3000
```

ROLE	EMAIL	PASSWORD	PRELOADED CONTEXT
Administrator	admin@studentms.internal	AdminStrongPass123!	Full institutional metrics, staff provisioning, audit records
Academic Staff	staff@studentms.internal	StaffTempPass123!	Restricted staff role, triggers first-login password change gate

6. Concurrent Execution & Troubleshooting Notes

Running Multiple Apps Simultaneously

By default, Next.js targets port 3000. To run multiple applications in parallel on distinct ports, specify the `PORT` environment variable:

```
# Terminal 1 (Expense Tracker)
cd expense_tracker && PORT=3000 npm run dev

# Terminal 2 (Job Portal)
cd job_portal_application && PORT=3001 npm run dev

# Terminal 3 (LMS)
cd learning_management_system && PORT=3002 npm run dev

# Terminal 4 (Student MS)
cd student_management_system && PORT=3003 npm run dev
```

Troubleshooting & Tips

- MongoDB Atlas IP Whitelist:** If connecting your own cluster, ensure IP access is enabled under *Atlas > Network Access > Add IP Address > 0.0.0.0/0 (Allow access from anywhere)*.
- Automated Testing:** Run `npm test` inside any project to execute the Vitest test suite against isolated in-memory databases (`mongodb-memory-server`).

